

PASSI

Progressi delle Aziende Sanitarie per la Salute in Italia

Continuous surveillance to describe health, behaviours and prevention among the adult population.

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6th Summer School in Social Statistics



Source: ISS, PASSI Surveillance, EpiCentro

The PASSI Data

Italian Behavioral and Risk Factors Surveillance System:

**Istituto Superiore di Sanità**
EpiCentro - L'epidemiologia per la sanità pubblica

Indice A-Z dei contenuti

Cerca.. 

Sorveglianza PASSI

[Home](#) | [EpiCentro](#)  

 **PASSI**

PASSI sur

veillance

edizione italiana

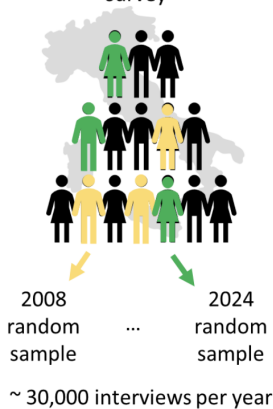
Progressi delle Aziende Sanitarie per la Salute in Italia (Passi)
The Italian behavioral risk factor surveillance system

Non-communicable chronic diseases cause the major part of the disease burden and have great economic impact, especially in countries with advanced economies. Behavioural risk factors play an important role in their development.

To plan, implement and evaluate preventive interventions on chronic diseases, timely information are needed not only on health status and behaviours of the population, but also on their knowledge and perceptions about healthy habits and lifestyle.



Cross-sectional repeated survey



Progress by Local Health Units towards a Healthier Italy

A public health surveillance system, active since 2008, that continuously collects self-reported information on health, lifestyles, risk factors and participation in prevention programmes among adults living in Italy.

- ▶ Reference model: *Behavioral Risk Factor Surveillance System*.
- ▶ Design: repeated cross-sectional sample surveys.
- ▶ Geographical levels: Local Health Unit, regional and national.
- ▶ Scientific and technical coordination: Italian National Institute of Health.

Source: ISS, PASSI Surveillance, EpiCentro

Surveillance That Turns Data into Action

How it works

- ▶ continuous data collection from representative samples of the adult population;
- ▶ standardised questionnaire and procedures;
- ▶ production of information within the ISS–Regions–Local Health Units network;
- ▶ integrated analysis at local, regional and national levels.

Why it is needed

- ▶ monitor risk factors and prevention needs;
- ▶ identify changes over time and geographical and social differences;
- ▶ produce timely and comparable indicators;
- ▶ inform programmes and policies and assess their reach.

PASSI is more than a data collection system.

It is a tool of the Italian National Health Service that links observation, prevention and public decision-making.

Source: ISS, PASSI Surveillance, EpiCentro

Who Is Interviewed and How?

Survey design

- ▶ residents aged 18–69, registered with a Local Health Unit and not institutionalised;
- ▶ monthly samples drawn from Local Health Unit registers;
- ▶ proportional stratification by sex and age group;
- ▶ telephone interviews throughout the year using a standardised questionnaire, conducted by trained staff;
- ▶ self-reported data rather than direct clinical measurements.

What does it produce?

- ▶ estimates of the **prevalence** of health behaviours, conditions and prevention uptake;
- ▶ comparisons across **population groups** and geographical areas;
- ▶ analyses of **changes over time**;
- ▶ identification of **social and health inequalities**.

Important

PASSI consists of a sequence of cross-sectional surveys: **it does not follow the same individuals over time.**

Source: ISS, PASSI Surveillance, EpiCentro

Response Rates and the Local PASSI Network



<http://www.epicentro.iss.it/passi/rapporto2011/metodi.asp>

Misure di monitoraggio della rilevazione Passi 2008-11 per Regione*

	Tasso di risposta	Tasso di rifiuto	Tasso di eleggibilità
Valle d'Aosta	85,5	11,0	95,9
Piemonte	78,1	13,5	95,8
Lombardia	89,1	8,1	94,9
Pa Bolzano	84,9	11,1	94,4
Pa Trento	89,3	6,2	95,0
Veneto	88,1	8,5	96,0
Friuli Venezia Giulia	89,0	6,3	95,2
Liguria	86,9	8,9	95,4
Emilia-Romagna	89,5	7,4	95,6
Toscana	83,3	11,4	96,1
Umbria	95,6	3,4	96,9
Marche	87,1	10,9	95,6
Lazio	87,0	7,8	93,4
Abruzzo	94,8	3,7	97,5
Molise	72,2	22,0	96,6
Campania	91,7	7,3	96,7
Puglia	88,3	8,2	92,8
Basilicata	83,7	14,4	94,4
Calabria	93,8	4,5	92,6
Sicilia	91,4	5,0	94,9
Sardegna	90,2	4,7	96,1
Pool di Asl	87,9	8,6	95,4

PASSI distinctive characteristics:

- Embedded in (run by) the NHS

- Local (interviews are administrated by Local Health Unit nurses)

The Italian PASSI example: total response rate **87.9**

Modular and adaptable structure

Behaviours

Physical activity, active mobility, diet, smoking, alcohol use, salt intake and weight status.

Health and well-being

Self-rated health, mental well-being and chronic conditions.

Prevention

Cancer screening, vaccinations and counselling from healthcare professionals.

Safety

Road and home safety, including driving under the influence.

Social determinants

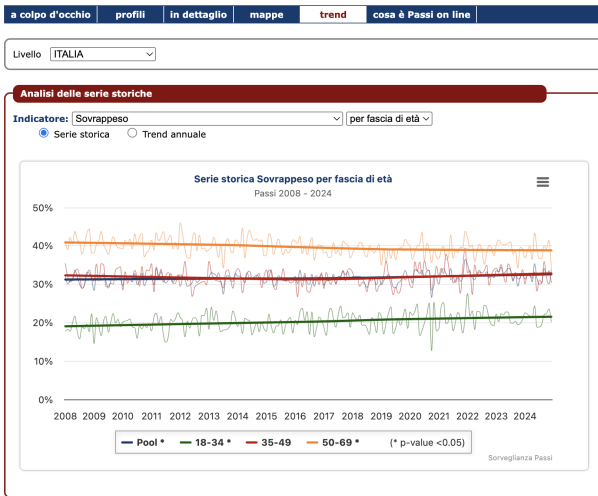
Education, employment, financial difficulties, citizenship and household composition.

Other topics

Optional regional modules addressing emerging topics.

Source: ISS, PASSI Surveillance, EpiCentro

Example of Data Use



<https://www.epicentro.iss.it/passi/dati/sovrappeso>

PASSI: A Wealth of Opportunities for Research

Applied questions

- ▶ **How do diseases, risk factors and multimorbidity evolve?**
[Epidemiology]
- ▶ **Who participates in prevention programmes, and which groups remain excluded?**
[Public health]
- ▶ **How do health and health behaviours differ by education, financial circumstances, occupation and citizenship?**
[Social inequalities]
- ▶ **How much do local context and environmental exposures matter?**
[Place and environment]

Methodological questions

- ▶ **How can temporal dynamics be reconstructed without following the same individuals?**
[Repeated cross-sectional data]
- ▶ **How should recall error, social desirability and response rounding be addressed?**
[Self-reported data]
- ▶ **How can multimorbidity, dependence among outcomes and spatio-temporal heterogeneity be modelled?**
[Multiple outcomes and contexts]
- ▶ **How can indicator comparability be ensured across years, geographical areas and questionnaire versions?**
[Data comparability]

Research Questions: Examples from the So.Sta. Group

- ▶ ...
- ▶ *Migrant health in Italy: a better health status difficult to maintain-country of origin and assimilation effects studied from the Italian risk factor surveillance data.*
Campostrini et al., *Popul. Health Metr.*, 2019 [[Migrant health and assimilation](#)]
- ▶ *When does morbidity start? An analysis of changes in morbidity between 2013 and 2019 in Italy.*
Pastore et al., *Stat. Methods Appl.*, 2023 [[Morbidity onset and ageing](#)]
- ▶ *Analysis of multimorbidity compression using a latent variable in a mixed mixture model.*
Andreella et al., *Popul. Health Metr.*, 2025 [[Multimorbidity compression](#)]
- ▶ *A Bayesian approach to model local and temporal heterogeneity in repeated cross-sectional health surveys.*
Stival et al., *JRSS-A*, 2026 [[Spatio-temporal health inequalities](#)]
- ▶ *Modeling Heterogeneity in Health Risk Behaviors With a Dynamic Mixture Model Informed by Textual Occupational Data.*
Schiavon et al., *Stat. Med.*, 2026 [[Occupational heterogeneity in health behaviours](#)]
- ▶ *Coarsening-adjusted estimators for finite-population indicators.*
Bertarelli and Gardini, Preprint, 2026 [[Survey estimation with coarsened data](#)]
- ▶ ...

The Summer School Dataset

Dataset

- ▶ **File:** `Passi20102019_reduced.csv`
- ▶ **Unit of analysis:** one row per respondent
- ▶ **Period:** 2010–2019
- ▶ **Geographical coverage:** Italy, with coverage varying over time
- ▶ **Sample size:** 100,000 respondents

Questionnaire and data dictionary

- ▶ **File:** `PASSI_tracciato_record_questionario_SSSS.xlsx`
- ▶ For each variable, check its name and label, question wording, eligible population, response codes, filters and skip patterns.
- ▶ Always verify its availability by survey year and geographical area.

Documentation

File: PASSI_tracciato_record_questionario_SSSS.xlsx

For each variable, check its name and label, question wording, eligible population, response codes and skip patterns.

The available sections are:

- ▶ **Section 1:** Health status and perceived quality of life
- ▶ **Section 2:** Physical activity
- ▶ **Section 3:** Smoking habits
- ▶ **Section 4:** Diet
- ▶ **Section 5:** Alcohol consumption
- ▶ **Section 12:** Mental health
- ▶ **Section 14:** Socio-demographic characteristics

Remember that

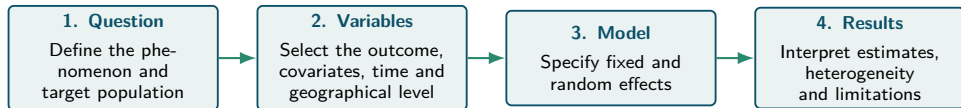
- ▶ The dataset includes only a selection of the variables collected through PASSI.
- ▶ The dataset has been restricted to a sample of 100,000 observations.
- ▶ Variable availability, coding, and missingness should be checked against the data dictionary.
- ▶ Data collection may not be consistent across time (e.g. Lombardia)
- ▶ Missing values, temporal discontinuities, and anomalous patterns should be carefully examined before conducting the analysis.

The Final Project

Objective

Working in small groups, formulate a research question using PASSI data and develop an analysis with the methods introduced during the Summer School, with particular attention to **Generalized Linear Mixed Models (GLMMs)**.

A simplified workflow



Example research question

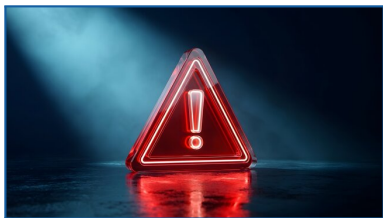
Do regional differences in the probability of smoking persist after accounting for sex and age? Did they change between 2010 and 2019?

A possible GLMM

$$\text{logit}[P(Y_{irt} = 1)] = \beta_0 + \mathbf{x}_{irt}^T \boldsymbol{\beta} + \beta_t + u_r$$

One final rule: keep the dataset within the Summer School

Explore the data, test your ideas and make the most of them!



Restricted use

The PASSI dataset provided for this course is intended **exclusively for teaching and project activities conducted within the Summer School.**

Please remember

The dataset must not be copied, shared, forwarded, uploaded or used outside the Summer School without explicit authorisation.

Enjoy working with the data and help us protect responsible access to them.